Spatial clustering of amyotrophic lateral sclerosis in Finland at place of birth and place of death.

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Previous evidence for spatial clustering of amyotrophic lateral sclerosis is inconclusive. Studies that have identified apparent clusters have often been based on a small number of cases, which means the results may have occurred by chance processes. Also, most studies have used the geographic location at the time of death as the basis for cluster detection, rather than exploring clusters at other points in the life cycle. In this study, the authors examine 1,000 cases of amyotrophic lateral sclerosis distributed throughout Finland who died between June 1985 and December 1995. Using a spatial-scan statistic, the authors examine whether there are significant clusters of the disease at both time of birth and time of death. Two significant, neighboring clusters were identified in southeast and south-central Finland at the time of death. A single significant cluster was identified in southeast Finland at the time of birth, closely matching one of the clusters identified at the time of death. These results are based on a large sample of cases, and they provide convincing evidence of spatial clustering of this condition. The results demonstrate also that, if the cluster analysis is conducted at different stages of the cases' life cycle, different conclusions about where potential risk factors may exist might result.